Report of Bitcoin Transaction Lab Assignment

Team - NodeSTAR

1) **Rudra Chitkara**
Roll Number: 240041031
Github: `https://github.com/Rudra-Codes`

2) **Aayush**
Roll Number: 240041001
Github: `https://github.com/sunno-bhaato`

3) **Tanish Yadav**
Roll Number: 240041036
Github: `https://github.com/tanpsi`

4) **Siddh Nema**
Roll Number: 240002070
Github: `https://github.com/SiddhaNema`

Part 1: Legacy Address Transactions (P2PKH)

Workflow

The transaction flow executes in two main stages based on the `transactions.py` code logic using the 'list_unspent', 'create_raw_tx', 'sign_raw_tx', and 'broadcast_tx' commands which map to RPC routines listunspent, createrawtransaction, signrawtransactionwithwallet, sendrawtransaction:

1. **Setup Phase:** We generate three base58-encoded legacy addresses: `A`, `B`, and `C`. Address `A` is then funded with **10 BTC** (as defined by `DEFAULT_FUND_A` in `config.py`) via the `sendtoaddress` RPC.

2. **Address A to B:** A raw transaction is created to send **9 BTC** (from `DEFAULT_TX_A_TO_B`) from `A` to `B`. We query `listunspent` for `A`'s addresses, extract UTXO(s) with at least 9 BTC, calculate the fee (default rate:10 satoshis/vbyte) using place holder fee and feedback loop and build inputs and outputs (9 BTC to B, the rest minus fees as change returned to A). The transaction is signed with A's private key (using the wallet) and broadcasted to the network. We then mine 1 block to confirm it. We also decode the signed transaction using decoderawtransaction which shows the scriptPubKey.

3. **Address B to C:** Another transaction sends **8 BTC** from `B` to `C`. We run `listunspent` for `B` to obtain the UTXO generated by the previous `A` $\rightarrow$ `B` transaction. This UTXO (bearing the exact `txid` from the last step) becomes the input for this new transaction. Similar fee calculations occur, we sign the transaction with B's private key, and broadcast it to the network. We also decode the signed transaction using decoderawtransaction.

Decoded Scripts

The decoded scripts representing the transaction details are provided below:

Decoded Script for A $\rightarrow$ B Transaction:

```
(.venv) [bofh@hole CS216-NodeSTAR-Bitcoin-Simulator]$ python main.py --type legacy --stage AtoB
================== Legacy Transactions ==================
INFO: Addresses
A: n3qDBhl89PeAYlYsn3wMFErkvcbGCAybXz
B: n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1
INFO: Funding A (1.0 BTC)
------------------- A -> B -------------------
INFO: Creating raw_tx (0.9 BTC)
INFO: Calculated fee: 2250 sat
INFO: Signing tx
INFO: Broadcasting tx
INFO: Signed tx hex:  020000000104dbfdd4ac4b7e99754dd3efd26588eb59bc7beffb64a99ad41a5b1cbdf330161000000006a47304402203546a3fe3743644d45c9c7da7384effa10ab95b22dcce74b0d2237a950b36c0c02207cc8e9c61ae379eb7a4f7b8ca739c667fca9b33e49079f9840a0e6
e353430d3f01210251a77a3cf55a03716739507475726dda3da8890181066f8b773928d3fe1c4336fdffffff02804a5d050000000001976a914f7e36f32aeae58f03bdf8da5362e96e5a7fe54688ac b68d900000000000001976a914f4c6af3c507377cb1ba91e74d362143ad43352ea88ac00000000
INFO: Decoding signed tx
{
    "txid": "9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28",
    "hash": "9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28",
    "version": 2,
    "size": 225,
    "vsize": 225,
    "weight": 900,
    "locktime": 0,
    "vin": [
        {
            "txid": "610133dfcbb1a541ad994ab6ffbec79bb58e5826fd3edd75997e4bacd4fddb04",
            "vout": 0,
            "scriptSig": {
                "asm": "304402203546a3fe3743644d45c9c7da7384effa10ab95b22dcce74b0d2237a950b36c0c02207cc8e9c61ae379eb7a4f7b8ca739c667fca9b33e49079f9840a0e6e353430d3f[ALL] 0251a77a3cf55a03716739507475726dda3da8890181066f8b773928d3fe1c4336",
                "hex": "473044022035 46a3fe3743644d45c9c7da7384effa10ab95b22dcce74b0d2237a950b36c0c02207cc8e9c61ae379eb7a4f7b8ca739c667fca9b33e49079f9840a0e6e353430d3f01210251a77a3cf55a03716739507475726dda3da8890181066f8b773928d3fe1c4336"
            },
            "sequence": 4294967293
        }
    ],
    "vout": [
        {
            "value": "0.90000000",
            "n": 0,
            "scriptPubKey": {
                "asm": "OP_DUP OP_HASH160 ff7e36f32aeae58f03bdf8da5362e96e5a7fe546 OP_EQUALVERIFY OP_CHECKSIG",
                "desc": "addr(n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1)#ffta6rvl",
                "hex": "76a914ff7e36f32aeae58f03bdf8da5362e96e5a7fe54688ac",
                "address": "n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1",
                "type": "pubkeyhash"
            }
        },
        {
            "value": "0.09997750",
            "n": 1,
            "scriptPubKey": {
                "asm": "OP_DUP OP_HASH160 f4c6af3c507377cb1ba91e74d362143ad43352ea OP_EQUALVERIFY OP_CHECKSIG",
                "desc": "addr(n3qDBhl89PeAYlYsn3wMFErkvcbGCAybXz)#7eg7jly0",
                "hex": "76a914f4c6af3c507377cb1ba91e74d362143ad43352ea88ac",
                "address": "n3qDBhl89PeAYlYsn3wMFErkvcbGCAybXz",
                "type": "pubkeyhash"
            }
        }
    ]
}
INFO: scriptPubKey:
OP_DUP OP_HASH160 ff7e36f32aeae58f03bdf8da5362e96e5a7fe546 OP_EQUALVERIFY OP_CHECKSIG
INFO: txid 9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28
```

Decoded Script for B → C Transaction:

```
(.venv) [bofh@hole CS216-NodeSTAR-Bitcoin-Simulator]$ python main.py --type legacy --stage BtoC
================== Legacy Transactions ==================
INFO: Addresses
B: n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1
C: mxbA1R61M1846qYg5fu5v3c2LzdsYNY4DF
------------------- B -> C -------------------
INFO: listunspent for B
[{'txid': '9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28', 'vout': 0, 'outpoint': COutPoint(lx('9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28'), 0), 'scriptPubKey': CScript([OP_DUP, OP_HASH160, x('ff7e36f32aeae58f03bdf8da5362e96e5a7fe
546'), OP_EQUALVERIFY, OP_CHECKSIG]), 'amount_sats': 90000000, 'confirmations': 102, 'address': 'n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1'}]
INFO: txid for UTXO of B: 9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28
INFO: Creating raw_tx (0.8 BTC)
INFO: Calculated fee: 2250 sat
INFO: Signing tx
INFO: Broadcasting tx
INFO: Signed tx hex:  020000000128be96948afb9ea54201ae97351d22eeca939654b9a3fd26bf1aa264f3318e9c000000006a47304402205a574a0f1541448e7b0841a9a57189c4fe59c7cac027038d140171d4443fe05302204579b3a5146ff210819510c2efa9590be676f84e2bc891333c9ab5b032e055d012103e5c1da485ec70183fc
512f664bb2e028d1a9c90234bf17acf482f36839573735fdffffff0200b4c4040000000001976a914bb45946e45d42908c2258bac57981c52f2823f1388acb68d900000000000001976a914ff7e36f32aeae58f03bdf8da5362e96e5a7fe54688ac00000000
INFO: Decoding signed tx
{
    "txid": "5172caaefc6c2abe936e6d0b3ab247a52ee1c8ccfc27e5d31d287c851aca223d",
    "hash": "5172caaefc6c2abe936e6d0b3ab247a52ee1c8ccfc27e5d31d287c851aca223d",
    "version": 2,
    "size": 225,
    "vsize": 225,
    "weight": 900,
    "locktime": 0,
    "vin": [
        {
            "txid": "9c8e31f364a21abf26fda3b9549693caee221d3597ae0142a59efb0a94968e28",
            "vout": 0,
            "scriptSig": {
                "asm": "304402205a574a0f1541448e7b0841a9a57189c4fe59c7cac027038d140171d4443fe05302204579b3a5146ff210819510c2efa9590be676f84e2bc891333c9ab5b032e055d0[ALL] 03e5c1da485ec70183fc512f664bb2e028d1a9c90234bf17acf482f36839573735",
                "hex": "473044022035a574a0f1541448e7b0841a9a57189c4fe59c7cac027038d140171d4443fe05302204579b3a5146ff210819510c2efa9590be676f84e2bc891333c9ab5b032e055d0012103e5c1da485ec70183fc512f664bb2e028d1a9c90234bf17acf482f36839573735"
            },
            "sequence": 4294967293
        }
    ],
    "vout": [
        {
            "value": "0.80000000",
            "n": 0,
            "scriptPubKey": {
                "asm": "OP_DUP OP_HASH160 bb45946e45d42908c2258bac57981c52f2823f13 OP_EQUALVERIFY OP_CHECKSIG",
                "desc": "addr(mxbA1R61M1846qYg5fu5v3c2LzdsYNY4DF)#m55e0n2c",
                "hex": "76a914bb45946e45d42908c2258bac57981c52f2823f1388ac",
                "address": "mxbA1R61M1846qYg5fu5v3c2LzdsYNY4DF",
                "type": "pubkeyhash"
            }
        },
        {
            "value": "0.09997750",
            "n": 1,
            "scriptPubKey": {
                "asm": "OP_DUP OP_HASH160 ff7e36f32aeae58f03bdf8da5362e96e5a7fe546 OP_EQUALVERIFY OP_CHECKSIG",
                "desc": "addr(n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1)#ffta6rvl",
                "hex": "76a914ff7e36f32aeae58f03bdf8da5362e96e5a7fe54688ac",
                "address": "n4osoX1fx6NHHdgA6NCwBEwtlhKCYfWWN1",
                "type": "pubkeyhash"
            }
        }
    ]
}
INFO: scriptSig:
304402205a574a0f1541448e7b0841a9a57189c4fe59c7cac027038d140171d4443fe05302204579b3a5146ff210819510c2efa9590be676f84e2bc891333c9ab5b032e055d0[ALL] 03e5c1da485ec70183fc512f664bb2e028d1a9c90234bf17acf482f36839573735
INFO: txid 5172caaefc6c2abe936e6d0b3ab247a52ee1c8ccfc27e5d31d287c851aca223d
```

Challenge and Response Scripts Structure

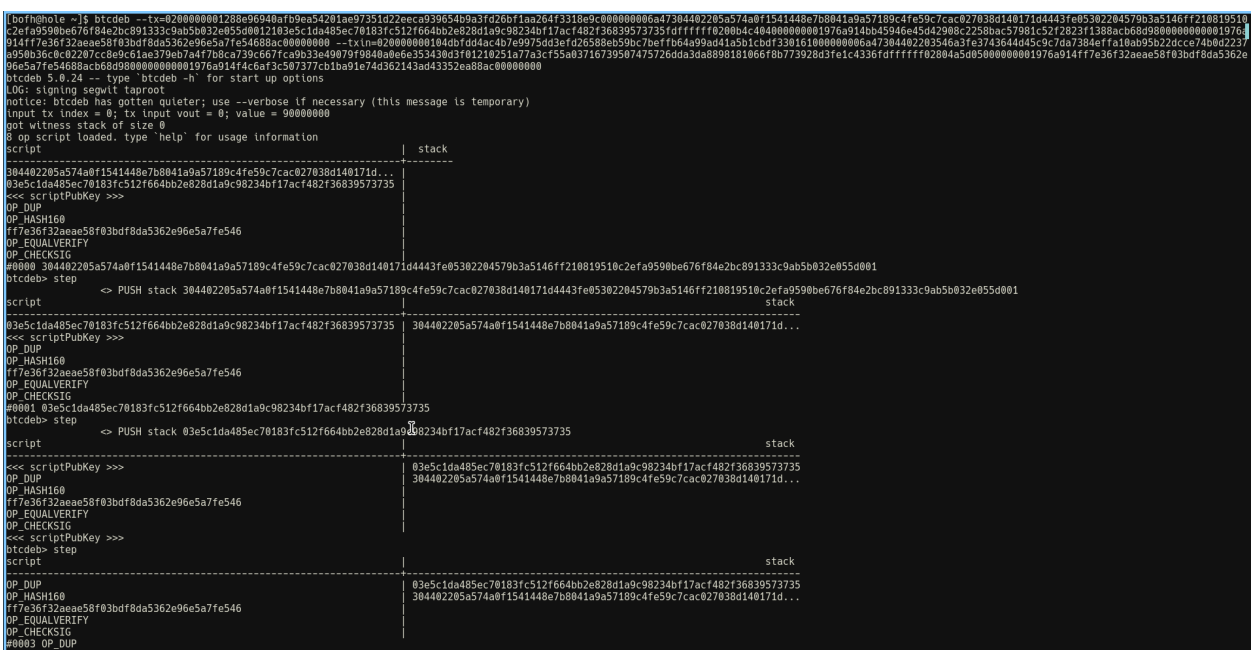
The validation of a Legacy P2PKH transaction involves combining the unlocking script (`scriptSig`) and the locking script (`scriptPubKey`):

- **Challenge Script (scriptPubKey):** `OP_DUP OP_HASH160 <PubkeyHash> OP_EQUALVERIFY OP_CHECKSIG`
- **Response Script (scriptSig):** `<Signature> <PublicKey>`

Validation Process: To spend a P2PKH UTXO, the spender must provide the full public key (which hashes to the `<PubkeyHash>`) and a valid signature. The network essentially prepends the `scriptSig` to the `scriptPubKey`. During validation:

1. `<Signature>` and `<PublicKey>` are pushed onto the stack.
2. `OP_DUP` duplicates the `PublicKey`.
3. `OP_HASH160` hashes the duplicated public key.
4. `<PubkeyHash>` is pushed, and `OP_EQUALVERIFY` checks if the generated hash equals the provided one. They match, popping both from the stack.
5. Finally, `OP_CHECKSIG` verifies the original signature against the original `PublicKey` and the transaction data. It evaluates to `True`, making the spend valid.

Screenshots Evidence for Btcdeb showing script execution in action



```
btcdeb> step
                <> PUSH stack 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
script                                                  |                                                                   stack
--------------------------------------------------------+-------------------------------------------------------------------
OP_HASH160                                              | 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
ff7e36f32aeae58f03bdf8da5362e96e5a7fe546                | 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
OP_EQUALVERIFY                                          | 304402205a574a0f1541448e7b8041a9a57189c4fe59c7cac02703Bd140171d...
OP_CHECKSIG                                             |
#0004 OP_HASH160
btcdeb> step
                <> POP  stack
                <> PUSH stack ff7e36f32aeae58f03bdf8da5362e96e5a7fe546
script                                                  |                                                                   stack
--------------------------------------------------------+-------------------------------------------------------------------
ff7e36f32aeae58f03bdf8da5362e96e5a7fe546                |                                   ff7e36f32aeae58f03bdf8da5362e96e5a7fe546
OP_EQUALVERIFY                                          | 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
OP_CHECKSIG                                             | 304402205a574a0f1541448e7b8041a9a57189c4fe59c7cac02703Bd140171d...
#0005 ff7e36f32aeae58f03bdf8da5362e96e5a7fe546
btcdeb> step
                <> PUSH stack ff7e36f32aeae58f03bdf8da5362e96e5a7fe546
script                                                  |                                                                   stack
--------------------------------------------------------+-------------------------------------------------------------------
OP_EQUALVERIFY                                          |                                   ff7e36f32aeae58f03bdf8da5362e96e5a7fe546
OP_CHECKSIG                                             |                                   ff7e36f32aeae58f03bdf8da5362e96e5a7fe546
                                                        | 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
                                                        | 304402205a574a0f1541448e7b8041a9a57189c4fe59c7cac02703Bd140171d...
#0006 OP_EQUALVERIFY
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
                <> POP  stack
script                                                  |                                                                   stack
--------------------------------------------------------+-------------------------------------------------------------------
OP_CHECKSIG                                             | 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
                                                        | 304402205a574a0f1541448e7b8041a9a57189c4fe59c7cac02703Bd140171d...
#0007 OP_CHECKSIG
btcdeb> step
EvalChecksig() sigversion=0
Eval Checksig Pre-Tapscript
GenericTransactionSignatureChecker::CheckECDSASignature(71 len sig, 33 len pubkey, sigversion=0)
  sig         = 304402205a574a0f1541448e7b8041a9a57189c4fe59c7cac02703Bd140171d4443fe05302204579b3a5146ff210819510c2efa9590be676f84e2bc891333c9ab5b032e055d001
  pub key     = 03e5c1da485ec70183fc512f664bb2e828d1a9c98234bf17acf482f36839573735
  script code = 76a914ff7e36f32aeae58f03bdf8da5362e96e5a7fe54688ac
  hash type   = 01 (SIGHASH_ALL)
SignatureHash(nIn=0, nHashType=01, amount=90000000)
- sigversion = SIGVERSION_BASE (non-segwit style)
 << txTo.vin[nInput=0].prevout = COutPoint(9c8e31f364, 0)
(SerializeScriptCode)
 << scriptCode.size()=25 - nCodeSeparators=0
 << script:76a914ff7e36f32aeae58f03bdf8da5362e96e5a7fe54688ac
 << txTo.vin[nInput].nSequence = 4294967293 [0xfffffffd]
  sighash     = 43bc550abd6c904e6e5790e20e00a481236cdf7dad0eef359f615e97dc3af629
  pubkey.VerifyECDSASignature(sig=304402205a574a0f1541448e7b8041a9a57189c4fe59c7cac02703Bd140171d4443fe05302204579b3a5146ff210819510c2efa9590be676f84e2bc891333c9ab5b032e055d0, sighash=43bc550abd6c904e6e5790e20e00a481236cdf7dad0eef359f615e
97dc3af629):
  result: success
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
script                                                  |                                                                   stack
--------------------------------------------------------+-------------------------------------------------------------------
                                                        |                                                                     01
```

Part 2: P2SH-SegWit Address Transactions

Workflow

The transaction flow follows a process identical to Part 1, modified for base58 P2SH address wrappers containing nested SegWit (P2WPKH) programs.

1. **Setup Phase:** Generate three P2SH-SegWit addresses (`A'`, `B'`, and `C'`), starting with "2" (on regtest). Address `A'` is funded with 10 BTC.
2. **Address A' to B':** We create a raw transaction to send 9 BTC to Address `B'` using the UTXOs for `A'`. We query listunspent for A's addresses, extract UTXO(s) with at least 9 BTC, calculate the fee (default rate:10 satoshis/vbyte) using place holder fee and feedback loop and build inputs and outputs (9 BTC to B, the rest minus fees as change returned to A). The transaction is signed with A's private key (using the wallet) and broadcasted to the network. We then mine 1 block to confirm it. We also decode the signed transaction using decoderawtransaction which shows the scriptPubKey.
3. **Address B' to C':** The transaction provides a UTXO locked to `B'`. We query the UTXOs for `B'`, isolate the exact `txid`, and create the next transaction. Upon signing, Bitcoin Core automatically places the SegWit signatures not in the `scriptSig`, but in the segregated `witness` data container.

Decoded Scripts

Decoded Script for A' $\rightarrow$ B' Transaction:

```
(.venv) [bofh@hole CS216-NodeSTAR-Bitcoin-Simulator]$ python main.py --type p2sh-segwit --stage AtoB
==================== P2SH-SegWit Transactions ====================
INFO: Addresses
A: 2NBxu4nTXJSBRSJsruP2oEteKMz9vAApHsY
B: 2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW
INFO: Funding A (1.0 BTC)
-------------------- A -> B --------------------
INFO: Creating raw_tx (0.9 BTC)
INFO: Calculated fee: 1660 sat
INFO: Signing tx
INFO: Broadcasting tx
INFO: Signed tx hex:  02000000000101766182633a6a1fb9808299af76a755e854c24639ad6f7B30170510cd4160dcc000000001716001 4ad494991db15779f9c9eb192f380af14ed403054fdffffff02084a5d050000000017a914799f5a0f673bee2d7c047abd7912f7b746c7f2918704909000000000017a914ac6b8fc01ab7c527566a
c73d47c857a2fc4dc063B7024738442280 1ac34cfef7f6f6091162529 0ede66b0eb2689147c626c8a4b35900a4d47a1ec3022070481b558b2d9eba3414b2984348a0044751bb4f465fb4f2025c7590f257502a01218307491ea7831accfaa29f44f8ca37c2312a0911d9f5ac844de0b256ef2507de7b4600000000
INFO: Decoding signed tx
{
    "txid": "248725c4ccc64ec8b708a8bf97b32dd5f2da299c6db57f802cdb29c16cabcdbc",
    "hash": "0b3c2a95ca8835edff73cafb13d6fa2f5686c81765cee70db0f09f78462a69b4",
    "version": 2,
    "size": 247,
    "vsize": 166,
    "weight": 661,
    "locktime": 0,
    "vin": [
        {
            "txid": "cc0d16d40c51700183f7d69a63244c855e756af79a290090fba1a6336282617 6",
            "vout": 0,
            "scriptSig": {
                "asm": "0014ad494991db15779f9c9eb192f380af14ed403054",
                "hex": "160014ad494991db15779f9c9eb192f380af14ed403054"
            },
            "txinwitness": [
                "3044022001ac34cfef7f6f6091162529 0ede66b0eb2689147c626c8a4b35900a4d47a1ec3022070481b558b2d9eba3414b2984348a0044751bb4f465fb4f2025c7590f257502a01",
                "0307491ea7831accfaa29f44f8ca37c2312a0911d9f5ac844de0b256ef2507de7b46"
            ],
            "sequence": 4294967293
        }
    ],
    "vout": [
        {
            "value": "0.90000000",
            "n": 0,
            "scriptPubKey": {
                "asm": "OP_HASH160 799f5a0f673bee2d7c047abd7912f7b746c7f291 OP_EQUAL",
                "desc": "addr(2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW)#alkwlv76",
                "hex": "a914799f5a0f673bee2d7c047abd7912f7b746c7f291 87",
                "address": "2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW",
                "type": "scripthash"
            }
        },
        {
            "value": "0.09998340",
            "n": 1,
            "scriptPubKey": {
                "asm": "OP_HASH160 ac6b8fc01ab7c527566ac73d47c857a2fc4dc063 OP_EQUAL",
                "desc": "addr(2N8xu4nTXJSBRSJsruP2oEteKMz9vAApHsY)#92t8hxst",
                "hex": "a914ac6b8fc01ab7c527566ac73d47c857a2fc4dc063 87",
                "address": "2N8xu4nTXJSBRSJsruP2oEteKMz9vAApHsY",
                "type": "scripthash"
            }
        }
    ]
}
INFO: scriptPubKey:
OP_HASH160 799f5a0f673bee2d7c047abd7912f7b746c7f291 OP_EQUAL
INFO: txid 248725c4ccc64ec8b708a8bf97b32dd5f2da299c6db57f802cdb29c16cabcdbc
```

Decoded Script for B' → C' Transaction:

```
(.venv) [bofh@hole CS216-NodeSTAR-Bitcoin-Simulator]$ python main.py --type p2sh-segwit --stage BtoC
==================== P2SH-SegWit Transactions ====================
INFO: Addresses
B: 2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW
C: 2NE53PBsc18D1fSgbV7quSTfL9EUDh51mqP
-------------------- B -> C --------------------
INFO: Unspent for B
[{'txid': '248725c4ccc64ec8b708a8bf97b32dd5f2da299c6db57f802cdb29c16cabcdbc', 'vout': 0, 'outpoint': COutPoint(lx('248725c4ccc64ec8b708a8bf97b32dd5f2da299c6db57f802cdb29c16cabcdbc'), 0), 'scriptPubKey': CScript([OP_HASH160, x('799f5a0f673bee2d7c047abd7912f7b746c7f291'), O
P_EQUAL]), 'amount_sats': 90000000, 'confirmations': 102, 'address': '2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW'}]
INFO: txid for UTXO of B: 248725c4ccc64ec8b708a8bf97b32dd5f2da299c6db57f802cdb29c16cabcdbc
INFO: Creating raw_tx (0.8 BTC)
INFO: Calculated fee: 1660 sat
INFO: Signing tx
INFO: Broadcasting tx
INFO: Signed tx hex:  02000000000101bccdab6cc129db2c807fb56d9c29daf2d52db397bfa800b7c84ec6ccc42507240000000017160014d6999a851cfb17343ad71b0f9fdfb6ec939829f9fdffffff0200bAc404800000000017a914e46da11012da0a29b37e096fceadced27155ffef0704909000000000017a914799f5a0f673bee2d7c04
7abd7912f7b746c7f2918702473044022022d1f1a59617ea94b9c2120d5a02fddd b423b8044cfa5162e9d8e3bb26e93dcd02205030cdcd16ca801fb99a617de61cfe04c25b06b3864529e3e02918dc4a8cdbbe0121824747 7b185fed00d0ea257c9feae10042fb98a3070dddd2eba6a23a2453d58b960000000
INFO: Decoding signed tx
{
    "txid": "d2992ab7f276e00d31508e06af5c9a7644d85c3999ea196b07ab1b506c61f13c",
    "hash": "4ddbf37497cb9abecff7666935ce6e8256311ba0dacff03e2e0462d341e2cc0d",
    "version": 2,
    "size": 247,
    "vsize": 166,
    "weight": 661,
    "locktime": 0,
    "vin": [
        {
            "txid": "248725c4ccc64ec8b708a8bf97b32dd5f2da299c6db57f802cdb29c16cabcdbc",
            "vout": 0,
            "scriptSig": {
                "asm": "0014d6999a851cfb17343ad71b0f9fdfb6ec939829f9",
                "hex": "160014d6999a851cfb17343ad71b0f9fdfb6ec939829f9"
            },
            "txinwitness": [
                "3044022022d1f1a59617ea94b9c2120d5a02fddd b423b8044cfa5162e9d8e3bb26e93dcd02205030cdcd16ca801fb99a617de61cfe04c25b06b3864529e3e02918dc4a8cdbbe01",
                "02474770185fed00d0ea257c9feae10042fb98a3070dddd2eba6a23a2453d58b96"
            ],
            "sequence": 4294967293
        }
    ],
    "vout": [
        {
            "value": "0.08000000",
            "n": 0,
            "scriptPubKey": {
                "asm": "OP_HASH160 e46da11012da0a29b37e096fceadced27155ffef OP_EQUAL",
                "desc": "addr(2NE53PBsc18D1fSgbV7quSTfL9EUDh51mqP)#6a38ae77",
                "hex": "a914e46da11012da0a29b37e096fceadced27155ffef87",
                "address": "2NE53PBsc18D1fSgbV7quSTfL9EUDh51mqP",
                "type": "scripthash"
            }
        },
        {
            "value": "0.09998340",
            "n": 1,
            "scriptPubKey": {
                "asm": "OP_HASH160 799f5a0f673bee2d7c047abd7912f7b746c7f291 OP_EQUAL",
                "desc": "addr(2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW)#alkwlv76",
                "hex": "a914799f5a0f673bee2d7c047abd7912f7b746c7f291 87",
                "address": "2N4LJeTCEEVPSzCuGdTPCsjoBKn7RTUx4dW",
                "type": "scripthash"
            }
        }
    ]
}
INFO: scrictSig:
0014d6999a851cfb17343ad71b0f9fdfb6ec939829f9
INFO: txid d2992ab7f276e00d31508e06af5c9a7644d85c3999ea196b07ab1b506c61f13c
```

Challenge and Response Scripts Structure

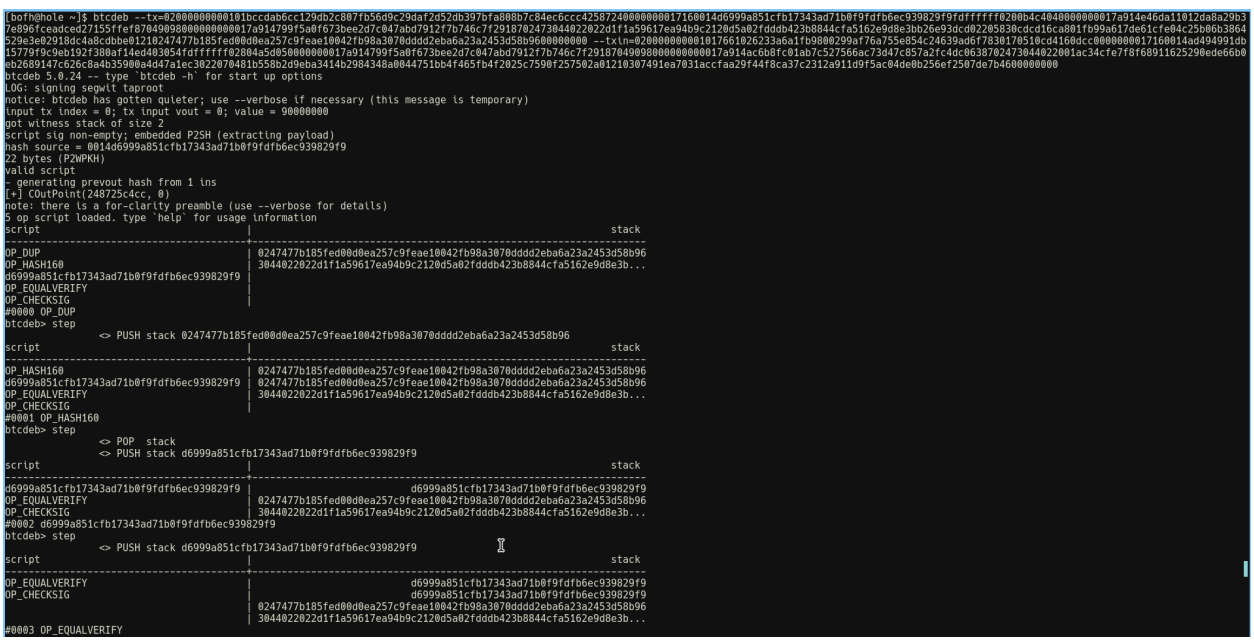
For nested P2SH-P2WPKH transactions, SegWit details are nested within a P2SH construct to maintain backward compatibility with older nodes:

- **Challenge Script (scriptPubKey):** `OP_HASH160 <Hash160(RedeemScript)> OP_EQUAL`
- **Response Script (scriptSig):** `<0> <20-byte-PubkeyHash>`
- **Witness Stack:** `<Signature> <PublicKey>`

Validation Process:

1. Old nodes view this as a standard P2SH check. They hash the provided `RedeemScript` (`0 <20-byte-PubkeyHash>`) from the `scriptSig` and ensure it matches the 20-byte hash in the `scriptPubKey`. They stop here, assuming it's valid.
2. Upgraded SegWit-aware nodes see that the supplied redeem script follows the very specific pattern of `0` followed by a 20-byte hash (the witness program).
3. This triggers the secondary validation step: retrieving the external `witness` stack. The execution context switches to evaluate it exactly like a native SegWit P2WPKH transaction, duplicating the public key, checking its hash against the 20-byte hash in the witness program, and evaluating `OP_CHECKSIG` using the provided signature!

Screenshots Evidence for Btcdeb showing script execution in action



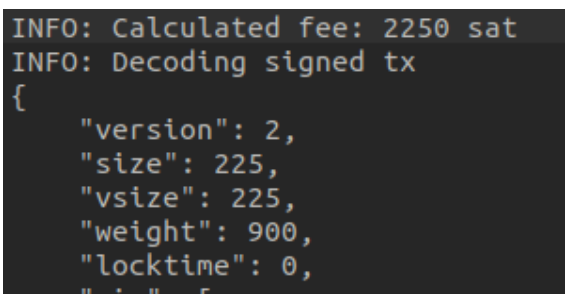
```
btcdeb> step
EvalChecksig() sigversion=1
Eval Checksig Pre-Tapscript
GenericTransactionSignatureChecker::CheckECDSASignature(71 len sig, 33 len pubkey, sigversion=1)
  sig         = 3044022022d1f1a59617ea94b9c2120d5a02fdddb423b8844cfa5162e9d8e3bb26e93dcd02205830cdcd16ca801fb99a617de61cfe04c25b06b3864529e3e02918dc4a8cdbbe01
  pub key     = 0247477b185fed00d0ea257c9feae10042fb98a3070dddd2eba6a23a2453d58b96
  script code = 76a914d6999a851cfb17343ad71b0f9fdfb6ec939829f988ac
  hash type   = 01 (SIGHASH_ALL)
SignatureHash(nIn=0, nHashType=01, amount=90000000)
- sigversion == SIGVERSION_WITNESS_V0
  sighash     = 44a83677451be8b4324c2327c1fed72c9a254ba2b7d6536b5575eab52f75120e
  pubkey.VerifyECDSASignature(sig=3044022022d1f1a59617ea94b9c2120d5a02fdddb423b8844cfa5162e9d8e3bb26e93dcd02205830cdcd16ca801fb99a617de61cfe04c25b06b3864529e3e02918dc4a8cdbbe, sighash=44a83677451be8b4324c2327c1f
ed72c9a254ba2b7d6536b5575eab52f75120e):
  result: success
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
script                                  |                                                       stack
----------------------------------------+-------------------------------------------------------------
                                        |                                                          01
btcdeb> ^C
```

Part 3: Analysis and Explanation

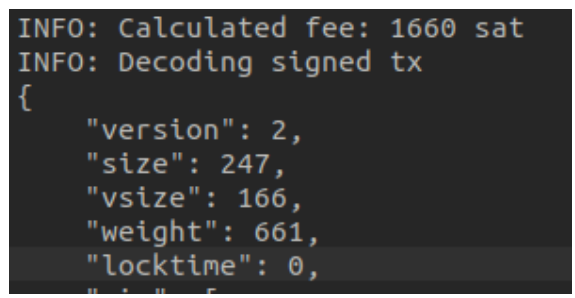
Size Comparison of P2PKH and P2SH-SegWit Transactions

By observing the differences between Part 1 and Part 2 decoded JSON objects (specifically the `size`, `vsize`, and `weight` fields), we highlight distinct space efficiencies:

- **Legacy P2PKH Size:** The signature script is embedded directly into the transaction input (in the `scriptSig`), weighing identically as the rest of the transaction. A typical 1-input 1-output P2PKH transaction is heavily penalized. It has a raw size and virtual size mapping of $1:1$ (as shown in image ~ 225 bytes translates to ~ 225 vBytes and 900 weight units).
- **P2SH-SegWit Size:** Witness data is segregated from the standard transaction tree structure. While P2SH-SegWit often produces a slightly higher absolute raw byte count because of the P2SH wrapper overhead (e.g., ~ 247 bytes), the *virtual size* (vBytes) and *weight* are significantly lower (e.g., ~ 166 vBytes and ~ 661 weight units).



(a) Legacy Size Info



(b) Segwit Size Info

Figure 1: Compare how virtual size and fees decreases in segwit as compared to legacy for same transactions size and fees

Benefits of SegWit

The SegWit (Segregated Witness BIP 141) architecture introduces wide-reaching benefits to the Bitcoin ecosystem:

1. **Lower Transaction Fees:** Users pay fees per virtual byte (vByte). Because witness data (signatures) are given a 75% discount (1 byte $=$ 1 weight unit $=$ 0.25 vBytes), SegWit transactions inherently incur drastically cheaper miner fees to occupy block space.
2. **Effective Block Size Increase:** This weight discounting model acts as a soft fork block size bump. A block filled exclusively with SegWit transactions could approximately hold 1.61x more data compared to legacy.

3. **Transaction Malleability Fix:** Because the signature data determines its own independent witness hash (‘wtxid‘) rather than the core transaction ID (‘txid‘), a third party altering a signature cannot mutate the ‘txid‘. This solidifies unconfirmed transactions, enabling reliable Layer-2 solutions like the Lightning Network.